

1 Chapter 6.1: Set Theory

Problem 5.13: Prove by Mathematical induction $\forall n \in \mathbb{Z}^+ \cup \{0\}, x, y \in \mathbb{Z}, x \neq y, x - y | x^n - y^n$

Proof Suppose $x, y \in \mathbb{Z} \vee x \neq y$. Let the predicate $P(n)$ be the statement $x - y | x^n - y^n$.

Basis Step Show that $P(0)$ is true. $P(0)$ is true because $x^0 - y^0 = 1 - 1$ and $x - y | 0$.

Inductive Step Show that $\forall k \in \mathbb{Z}^+ \cup \{0\}, P(k) \rightarrow P(k + 1)$

Let $k \in \mathbb{Z}^+ \cup \{0\}$ and suppose $x - y | x^k - y^k$. We must show that

$$x - y | x^{k+1} - y^{k+1}$$

. By rewriting the inductive hypothesis using the definition of divisibility, we have:

$$\exists r \in \mathbb{Z}, x^k - y^k = (x - y)r$$

Then:

$$\begin{aligned} & x^{k+1} - y^{k+1} \\ = & \\ & x^{k+1} - xy^k + xy^k - y^{k+1} \\ = & \\ & x(x^k - y^k) + y^k(x - y) \\ = & \\ & x(x - y)r + y^k(x - y) \\ = & \\ & (x - y)[xr + y^k] \end{aligned}$$

Now

$$x(r + y^k) \in \mathbb{Z}$$

because products and sums of integers are integers. Thus,

$$x - y | x^{k+1} - y^{k+1}$$

.

Let the set

$$A = \{x \in \mathbb{Z} | x = 5a + 2, a \in \mathbb{Z}\}$$

$$B = \{y \in \mathbb{Z} | y = 10b - 3, b \in \mathbb{Z}\}$$

$$C = \{z \in \mathbb{Z} | z = 10c + 7, z \in \mathbb{Z}\}$$

A. Is $A \subset B$? . No, because $2 = 5 * 0 + 2$, but $2 \notin B$ because if $2 \in B$ then $\exists b \in \mathbb{Z}$ such that $2 = 10b - 3$. Then $10b = 5, b = \frac{1}{2}$ which is not an integer. Counterexample found.

B. Is $B \subset A$? . Yes. Proof: Suppose $y \in B$. [*We must show that $y \in A$. By definition of A, this means that we must show $y = 5(\text{some int}) + 2$]. By definition of B, $\exists b \in \mathbb{Z}, y = 10b - 3$. Let $a = 2b - 1$. Then $a \in \mathbb{Z}$ and $5a + 2 =$

$$5(2b - 1) + 2 = 10b - 5 + 2 = 10b - 3 = y$$

To prove $A = B$, prove both are subsets of each other Let the universal set be $\mathbb{R}$

$$A \cup B = \{x \in \mathbb{R} | 6 < x \leq 8\}$$

$$A \cap B = \{x \in \mathbb{R} | -1 < x \leq 0\}$$

$$A^c = \{x \in \mathbb{R} | -3 > x \text{ or } x > 0\}$$

$$A \cap C = \emptyset$$

*Draw the venn diagram to describe $A \cap B \neq \emptyset, B \cap C = \emptyset, A \cap C = \emptyset, A \not\subset B, C \not\subset B$ *

Questions: Is $0 \in \emptyset$? No $\emptyset \neq \{\emptyset\}$? Yes. $\emptyset \in \{\emptyset\}$ $\emptyset \notin \emptyset$

EXAM QUESTION

$$\forall i \in \mathbb{Z}^+, R_i = \{x \in \mathbb{R} | 1 \leq x \leq 1 + \frac{1}{i}\} = [1, 1 + \frac{1}{i}]$$

$$\cup_{i=1}^4 R_i = R_1 \cup R_2 \cup R_3 \cup R_4 = \cup [1, 1 + \frac{1}{1}] \cup [1, 1 + \frac{1}{2}] \cup [1, 1 + \frac{1}{3}] \cup [1, 1 + \frac{1}{4}] = [1, 2]$$

$$\cap_{i=1}^4 R_i = R_1 \cap R_2 \cap R_3 \cap R_4 = \cap [1, 1 + \frac{1}{1}] \cap [1, 1 + \frac{1}{2}] \cap [1, 1 + \frac{1}{3}] \cap [1, 1 + \frac{1}{4}] = [1, 1 + \frac{1}{4}]$$

Is $\{\{w, x, v\}, \{w, y, q\}, \{p, z\}\}$ a partition of $\{p, q, u, v, w, x, y, z\}$? Yes.